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Session 3 — Introduction to CSS and Styling

Objective: Students understand CSS syntax, the three ways to apply styles, and common properties.

1. Key Concepts

What is CSS?

CSS (Cascading Style Sheets) controls the *appearance* of HTML elements: colours, fonts, sizes, spacing, and layout. HTML defines *what* is on the page; CSS defines *how it looks*.

CSS Syntax

A CSS rule is made of three parts:

- **Selector:** which element(s) to target.
- **Property:** which visual aspect to change.
- **Value:** what to set the property to.

```
selector {  
    property: value;    /* one declaration */  
    property: value;    /* rules can have many declarations */  
}
```

Three Ways to Apply CSS

Concept	Meaning
Inline	style='...' attribute directly on the HTML tag. Overrides all others.
Internal	Inside a <style> tag in the <head> section. Applies to that page only.
External	In a separate .css file linked with <link rel='stylesheet' href='style.css'>.

External CSS is the preferred approach for real projects — it keeps style and structure separate and one .css file can style an entire website.

Essential CSS Properties

Concept	Meaning
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color	Text colour.
background-color	Background fill of an element.
font-size	Text size. Always specify a unit: px, em, or %.
font-family	Typeface. Provide a fallback: Arial, sans-serif.
text-align	Horizontal text alignment: left, center, right, justify.
text-decoration	Underline, overline, line-through, or none.
font-weight	Text thickness: normal or bold (or 100–900).

2. Guided Examples

Example A — Internal CSS

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      body { background-color: #f0f4f8; font-family: Arial, sans-serif; }
      h1   { color: #1A3A5C; text-align: center; font-size: 28px; }
      p    { color: #333333; font-size: 14px; text-align: justify; }
    </style>
  </head>
  <body>
    <h1>Welcome</h1>
    <p>This page uses internal CSS.</p>
  </body>
</html>
```

Example B — External CSS (style.css)

```
/* style.css */
h2 {
  color: darkblue;
  text-decoration: underline;
}
a {
  color: orangered;
  font-weight: bold;
}
```

3. Exercises

Exercise 1 — Decompose and Analyse CSS Rules

For each CSS rule below, fill in the table with the **selector**, **property**, and **value**. Then write a one-sentence description of the visual effect.

```
/* Rule A */ p { color: #2E4057; font-size: 16px; }
/* Rule B */ h1 { text-align: center; background-color: navy; color: white;
}
/* Rule C */ body { font-family: Georgia, serif; line-height: 1.6; }
```

Rule	Selector	Properties & Values	Visual effect
A			
B			
C			

Exercise 2 — Write CSS from a Description

Write the CSS rules (using internal CSS inside a <style> block) for a web page that must look like this:

- The page background is light grey (#f5f5f5).
- All headings (<h1> and <h2>) are dark red (#8B0000), centred, and use Arial font.
- Paragraphs have font size 14px, justified text, and a line-height of 1.5.
- All links are green, bold, and have no underline (text-decoration: none).

Write your CSS below:

Exercise 3 — Three Methods — Compare and Choose

For each scenario below, state which CSS method (inline, internal, or external) is most appropriate and explain why in one sentence.

Scenario	Best method	Reason
You want a quick red text on one specific word.		
A website has 20 pages, all sharing the same colours and fonts.		
You are writing a single-page prototype and want styles near the HTML.		
You need to override one style for exactly one element on one page.		

Exercise 4 — Predict the Rendered Output

Read the complete HTML+CSS page below. Describe in words exactly how it would look in a browser. Be precise about colours, sizes, and alignment.

```
<style>
  body { background-color: black; color: white; font-family: Courier; }
  h1   { color: gold; text-align: center; font-size: 36px; }
  p    { font-size: 12px; text-decoration: underline; }
</style>
<body>
  <h1>Hacker Zone</h1>
  <p>All systems operational.</p>
  <p>Access granted.</p>
</body>
```

Description of how the page looks:

Exercise 5 — Debug the CSS

The following CSS block is meant to style an article page but contains **5 errors**. Identify and fix each one.

```
body {  
    background colour: lightyellow;  
    font-family: Arial  
}  
  
h1 {  
    color: darkblue;  
    font-size: 24;  
    text-align: centre;  
}  
  
p {  
    colour: #333;  
    line-height 1.5;  
}
```

Error 1: _____

Error 2: _____

Error 3: _____

Error 4: _____

Error 5: _____

4. Session Recap

CSS (Cascading Style Sheets) separates content from presentation. A CSS rule always has a selector, one or more properties, and their values. CSS can be applied inline (on one tag), internally (in a <style> block), or externally (in a .css file). Key properties include color, background-color, font-size, font-family, and text-align. Next session, we explore the box model and layout.